

Chapter 2

Literature Review:

This chapter reviews the theoretical foundation for studying quantum correlations in integrated photonic circuits. It opens with an overview of Continuous-Variable (CV) quantum states, focusing particularly on Gaussian states, whose properties are fully captured through the covariance matrix Σ derived from the field quadratures. The discussion then moves to the physical platform of optical waveguides, describing how they guide light and how evanescent coupling naturally facilitates interaction between neighbouring modes. Special attention is given to the role of loss and decoherence: ordinary optical attenuation (α) couples the system to the vacuum and manifests as quantum decay (Γ), limiting the preservation of quantum information. The chapter proceeds to outline the two key quantum resources explored in this work: quantum steering, described through the Gaussian steering measure $\mathcal{S}$, and entanglement, characterized by the logarithmic negativity E_N . These elements collectively establish the groundwork for studying the evolution of these resources within coupled waveguide architectures. This chapter concludes by discussing the essential applications of entanglement, steering and

waveguides.

2.1 Quantum states of light

2.1.1 Continious variable states

In quantum mechanics, a Continious variable state is a quantum state where the physical observables i,e momentum and position vary continiously unlike discrete variable state where observables can take only discrete values. These system are illustrated with in an infinite-dimensional Hilbert space [?]. In these system the principal variables are the quadrature operators which obey specific commutation relations that form the foundation of the system's quantum structure [?]. Continious variable quantum states are classified into two catogories: Gaussian and non Guassian states [?].

2.1.2 Guassian states

In a continious variable system quantum system, a Guassian state is a special kind of quantum state whose fourier representation has a guassian form in phase space. These states are completely described by their first and second moments. The quadrature operators are

$$Q = \frac{1}{\sqrt{2}}(a + a^\dagger), \quad R = \frac{i}{\sqrt{2}}(a^\dagger - a), \quad (2.1)$$

which satisfy the canonical commutation relation

$$[Q, R] = i. \quad [?] \quad (2.2)$$

The Wigner function of a Gaussian state is

$$W(\mathbf{z}) = \frac{1}{(2\pi)^m \sqrt{\det \Sigma}} \exp \left[-\frac{1}{2} (\mathbf{z} - \bar{\mathbf{z}})^T \Sigma^{-1} (\mathbf{z} - \bar{\mathbf{z}}) \right]. \quad (2.3)$$

Here,

$$\mathbf{z} = (Q_1, R_1, \dots, Q_m, R_m)^T$$

is the phase-space vector,

$$\bar{\mathbf{z}} = \langle \hat{\mathbf{z}} \rangle$$

is the displacement vector, and

$$\Sigma_{ij} = \frac{1}{2} \langle \hat{z}_i \hat{z}_j + \hat{z}_j \hat{z}_i \rangle - \langle \hat{z}_i \rangle \langle \hat{z}_j \rangle$$

is the covariance matrix.

All physical properties of the gaussian states such as purity, entropy, steering and entanglement can be determined directly from the covariance matrix without full density operator because of the gaussian form of weigner function. typical examples of gaussian states include the coherent states, squeezed states, thermal states and vaccume state [?, ?].

2.2 Covariance matrix

In continuous-variable quantum systems, the degrees of freedom of the bosonic modes are represented by the quadrature operators $\hat{Q}_k$ and $\hat{R}_k$. The phase-space vector $\hat{\mathbf{z}}$ is defined as

$$\hat{\mathbf{z}} = (\hat{Q}_1, \hat{P}_1, \dots, \hat{Q}_m, \hat{R}_m)^T. \quad (2.4)$$

The covariance matrix Σ for an n -mode quantum state is defined as

$$\Sigma_{ij} = \frac{1}{2} \langle \hat{z}_i \hat{z}_j + \hat{z}_j \hat{z}_i \rangle - \langle \hat{z}_i \rangle \langle \hat{z}_j \rangle. \quad (2.5)$$

Here, Σ is a real, symmetric matrix that must satisfy the quantum uncertainty principle in order to represent a physical Gaussian state [?].

2.3 Optical waveguides as Quantum systems

An Optical waveguide is a structure that guide light along a desired path. It consists of thin, high refractive index core layer positioned between low refractive index cladding materials and the light is trapped within the core through total internal refrelction. Photons passes through these waveguides in particular patterns known as modes [?].

2.3.1 Evanscent overlap/coupled mode theory

When light passes through a waveguide system, the electric feild is mostly trapped in the high refractive index core. Yet, a part of the feild can penetrate into the surround-

ing lower-index cladding. We call this decaying field as the evanescent field. If we place another waveguide in this region i.e the evanescent field reach typically less than 1-2 wavelengths away light can tunnel between these two waveguides. We call this transfer of energy through coupling of evanescent fields as evanescent coupling. The coupling strength between the waveguides depends on the separation distance between the waveguides, refractive index and mode shapes [?].

2.3.2 loss and decoherence in real waveguide systems

In real optical waveguide system, the performance is undermined by two processes: classical optical power loss (attenuation α) and quantum information loss (decoherence, Γ). Exponential decay of optical intensity as light propagates is called attenuation while decoherence is the process in which a quantum system loses its correlation properties i.e superposition or entanglement due to interaction with the environment, results in losing the necessary phase relationship and limiting visibility of interference [?]. In quantum photonics the two problems are interconnected, classical optical loss (α) act as primary source of quantum decoherence. When a photon is lost from the waveguide system, its quantum state becomes coupled to unobserved vacuum port which results in transferring quantum information to the environment irreversibly and thus decoheres the entanglement and other nonclassical quantum resources. This loss mechanism is as an amplitude damping channel [?].

2.3.3 Relevance of optical waveguides for quantum circuits/photonics

Optical waveguides are very essential in quantum photonics. They made it possible for the researchers to move from large laboratory setups to small chips that in near future can be used in quantum technologies. These systems are very convenient and easy to reproduce because everything is constructed on a chip, enabling reliable implementation of quantum logic gates such as CNOT, Toffoli and Fredkin gate on a chip. These quantum gates are essential for building quantum computers, secure communication and quantum sensors. For this purpose low loss photonic materials such as silicon, silicon nitride, and lithium niobate are very essential which plays critical role in maintaining error resilient quantum operations in integrated photonic circuits. Their useful optical and nonlinear properties allows optimized generation, manipulation, and transmission of quantum states with very low propagation loss. When a photon sources, beam splitters, phase shifters, and detectors are built together on a chip made of waveguides in-

stead of using bulky mirrors and lenses on an optical table, the resulting circuit performs functions like quantum logic gates and routing networks. Through this integration optical waveguides becomes active part of computation itself. They not only guiding light, they are performing logical operations of quantum computation. The physical behaviour of light inside the waveguide such as interference, coupling and phase control directly determines how quantum information is processed. which means a network of interconnected waveguides behaves as quantum logic processor itself [?, ?].

2.4 Steering in continuous variable

Quantum steering is a unique quantum correlation first identified in the context of the 1935 argument presented by Einstein, Podolsky, and Rosen (EPR) regarding the completeness of quantum mechanics. Erwin Schrödinger observed that one party (Alice) could influence, or "steer," the wave function of a distant party (Bob) merely by selecting a suitable local measurement, a phenomenon he considered "magic" [?]. In contemporary quantum information theory, quantum steering is formally defined by the impossibility of describing the set of conditional quantum states observed by Bob (known as the steering assemblage) using a Local Hidden State (LHS) model [?, ?]. An LHS model posits that Bob's quantum state originated from a preexisting hidden state, and Alice's measurement only provides classical information about that hidden state's distribution. Steering holds a defined position in the strict hierarchy of quantum correlations, situated between entanglement and Bell nonlocality. All steerable states are necessarily entangled, and all Bell nonlocal states are necessarily steerable [?]. The primary practical significance of steering lies in establishing secure quantum communication and verifying entanglement in a one-sided device-independent scenario, where Bob's measurement apparatus is assumed to be trusted (characterized), but Alice's is not [?, ?]. In continuous-variable (CV) systems, which are typically realized using optical fields (quadratures X and P), the detection of steering is often formalized using criteria derived from the Heisenberg uncertainty relation for conditional measurements. The steering criterion for a state shared between A and B is met if the product of the minimum inferred variances for Bob's position (X_B) and momentum (P_B) quadratures, based on Alice's measurements, is less than the quantum limit: [?, ?]

$$V_{\text{inf}}(X_B) V_{\text{inf}}(P_B) < 1. \quad (2.6)$$

For CV Gaussian states where both the states and measurements are Gaussian, this EPR criterion is both necessary and sufficient to detect bipartite steering[?]. Remarkably, a bipartite Gaussian state is unsteerable with Gaussian measurements if and only if a specific matrix inequality involving the covariance matrix V_{AB} is satisfied:

$$V_{AB} + i(0_A \oplus \Omega_B) \geq 0. \quad (2.7)$$

If the state is steerable with Gaussian measurements, it is also steerable with a certain pair of canonical quadratures [?].

2.5 Quantification of Quantum Steering

2.5.1 Gaussian Steering Measure

The Gaussian $A \rightarrow B$ steerability $\mathcal{S}_{A \rightarrow B}(\sigma_{AB})$ is defined based on the symplectic eigenvalues $\{\bar{\nu}_j^B\}$ of the Schur complement matrix M_{σ_B} , calculated from the state's covariance matrix σ_{AB} :

$$\mathcal{S}_{A \rightarrow B}(\sigma_{AB}) := \max \left\{ 0, - \sum_{\bar{\nu}_j^B < 1} \ln(\bar{\nu}_j^B) \right\}. \quad (2.8)$$

For the specific case of two-mode Gaussian states (where the steered party has only one mode), the measure simplifies to the Rényi-2 coherent information:

$$\mathcal{S}_{A \rightarrow B}(\sigma_{AB}) = \max \left\{ 0, \frac{1}{2} \ln \left(\frac{\det \sigma_A}{\det \sigma_{AB}} \right) \right\} = \max \{ 0, H_2(\sigma_A) - H_2(\sigma_{AB}) \}, \quad (2.9)$$

where $H_2(\sigma)$ denotes the Rényi-2 entropy.

Properties and Hierarchy: The Gaussian steering measure satisfies several desirable mathematical properties: it is invariant under local unitaries, convex, additive, and monotonically non-increasing under quantum operations performed on the untrusted party (Alice). It vanishes if and only if the state is nonsteerable by Gaussian measurements. Moreover, it respects the hierarchy of correlations:

$$\mathcal{S}_{A \rightarrow B}(\sigma_{AB}) \leq \mathcal{E}_2(\sigma_{AB}), \quad (2.10)$$

where $\mathcal{E}_2$ denotes the Gaussian Rényi-2 measure of entanglement. For pure states,

$$\mathcal{S}_{A \rightarrow B}(\sigma_{AB}^p) = \mathcal{E}_2(\sigma_{AB}^p). \quad (2.11)$$

Operational Meaning: For two-mode Gaussian states, Gaussian steerability has a direct operational interpretation in one-sided device-independent QKD:

$$\mathcal{K} \geq \max \{0, \mathcal{S}_{B \rightarrow A}(\sigma_{AB}) + \ln 2 - 1\}, \quad (2.12)$$

where $\mathcal{K}$ is the guaranteed secure key rate[?].

2.6 Entanglement in continious variable systems

Entanglement is fundamentally recognized as an intrinsic feature of quantum theory and serves as the primary resource underlying the majority of quantum information science and related technologies [?]. Originating from conceptual structures identified by Einstein, Podolsky, and Rosen (EPR) and Schrödinger, entanglement describes nonlocal correlations within composite quantum systems that cannot be replicated by classical systems [?]. It describes a physical phenomenon where two or more particles exhibit correlations that cannot be explained classically, remaining correlated regardless of the distance separating them [?]. Historically viewed as a "spooky" phenomenon, entanglement is now engineered to facilitate essential quantum protocols, including quantum computing and quantum teleportation in CV systems, associated with infinite-dimensional Hilbert spaces, offer the advantage of enabling deterministic and unconditional quantum operations, which is often a limitation in DV schemes due to reliance on imperfect single-photon generation and detection. In implementing quantum information protocols, information can be encoded using either discrete variables (DVs), such as the discrete levels of a single photon, or continuous variables (CVs), typically utilizing the quadrature phase amplitude of the electromagnetic field [?]. Within the CV framework, the particularly relevant family of Gaussian states is the primary focus of entanglement research [?]. The systematic effort to determine the level or degree of entanglement present in a quantum system is known as entanglement quantification. Quantification is critical because the measured degree of entanglement directly influences the performance of quantum information tasks[?].

2.6.1 Entanglement Quantification Measures

To quantitatively assess the entanglement generated and sustained within a system, researchers employ specific measures. The primary metric used in the studies reviewed is the logarithmic negativity E_N .

E_N is defined as

$$E_N(t) = \log_2 \|\rho^T\|, \quad (2.13)$$

where ρ^T denotes the partial transpose of the density matrix ρ , and $\|\cdot\|$ represents the trace norm. A non-zero value for E_N confirms that the state is entangled. For Gaussian states, such as squeezed states, the measure of entanglement, E_N , is determined by the symplectic eigenvalues ($\tilde{\nu}$) of the covariance matrix (σ). The state is entangled if the smallest symplectic eigenvalue, $\tilde{\nu}_<$, satisfies $\tilde{\nu}_< < \frac{1}{2}$. Logarithmic negativity in this context is given by

$$E_N(t) = \max[0, -\ln(2\tilde{\nu}_<(t))], \quad (2.14)$$

and serves as an upper bound to the distillable entanglement[?].

2.7 loss and decoherence

Decoherence has become a prominent subject in modern physics, motivated by the desire to understand the border between the quantum and classical worlds, and by the destructive role it plays in quantum information processing [?]. Fundamentally, decoherence is defined as the irreversible, uncontrollable, and persistent formation of quantum correlations (entanglement) between a system and its environment [?, ?]. Decoherence has attracted significant attention due to its role in explaining the boundary between the quantum and classical domains, and its damaging effects on quantum information processing technology. The key mechanism is that the environment effectively monitors certain system observables, which leads to the destruction of coherence between non-preferred states. This monitoring process compels the selection of pointer states, which are robust and stable against environmental interaction. These pointer states retain correlations with the surrounding universe despite being immersed in the environment. The resulting rapid conversion of quantum superpositions into mixtures explains the apparent absence of macroscopic superposition states, often referred to as "Schrödinger cat" states. The selection of these preferred states is termed Environment-Induced Superselection (Einselection). Einselection imposes an effective prohibition on the vast major-

ity of the system's Hilbert space, limiting it to a smaller classical domain composed of pointer states [?].

2.7.1 Loss (Dissipation) and Pure Decoherence

Decoherence is usually accompanied by dissipation, which refers to the net exchange of energy with the environment [?]. However, specific theoretical scenarios concentrate on pure decoherence (also known as dephasing), where the exchange of energy is negligible. Pure decoherence is particularly relevant for explaining why macroscopic superpositions are absent, as this explanation should primarily depend on the distinguishability of the states, rather than on their energy difference [?]. Pure decoherence occurs primarily when the system Hamiltonian (H_S) commutes approximately with the system-bath interaction Hamiltonian (H_{int}), or when the initial states of the system evolve very slowly under H_S relative to the decoherence timescale. Pure decoherence is considered the main component in theories aiming to explain the apparent absence of superpositions of macroscopically distinguishable states (e.g., "Schrödinger cat" states) [?]. It is important to distinguish the roles of these phenomena: while relaxation and noise (loss) are caused by the environment perturbing the system, decoherence and einselection occur because the system perturbs the environment [?]. A crucial aspect of decoherence, particularly for macroscopic systems, is the time scale separation between coherence loss and energy loss (relaxation). The rate of decoherence is often dependent on the "distance" between the superposed robust states. For states that are macroscopically distinct, the dimensionless distance D can be enormous (e.g., 10^{20}), leading to incredibly fast decoherence times, even if the system's energy relaxation rate is minimal [?].

2.7.2 Loss mechanism in waveguides

Loss (γ) in waveguides arises from the loss in the material of the waveguide structure itself. Since waveguides are often designed to be identical, the loss rate for the field modes (a and b) is typically considered the same. Loss is theoretically modeled within the framework of system–reservoir interaction using a quantum-Liouville equation of motion. This evolution equation for the density operator (ρ) includes a Hamiltonian term for unitary coupling and a loss term ($L\rho$) to account for dissipation:

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + L\rho, \quad (2.15)$$

where the loss operator $L\rho$ specifically accounts for the decay rates of the modes a and b [?].

2.8 Applications and motivations

2.8.1 Applications of Quantum Entanglement

Quantum entanglement is a foundational feature of quantum information theory and quantum mechanics, describing highly correlated particles that remain connected regardless of the distance separating them [?]. This phenomenon is essential for demonstrating the theoretical and algorithmic superiority of a quantum computer over its classical counterparts [?, ?]. Entanglement manipulation is a critical component of quantum computing, leading to increased capabilities [?], and is leveraged in core quantum information processing applications such as quantum cryptography, quantum teleportation, and quantum computing [?]. Quantifying entanglement is crucial for establishing the intactness and depth of the correlation level, which directly influences the performance of quantum information tasks. Quantification methods include concurrence, negativity (including logarithmic negativity), Schmidt decomposition, and entanglement of formation [?]. For bipartite Gaussian states, the logarithmic negativity is often used to quantify entanglement. Within waveguide architectures, the sustainability of generated entanglement is very important for effective use in quantum communication and computation tasks, and entanglement dynamics are studied quantitatively using logarithmic negativity for input states like squeezed states and photon number states [?]. Highly entangled states are critical for performing complex quantum information processing and implementing error-correcting codes [?].

2.8.2 Applications of Quantum Steering

Quantum steering (QS), a type of quantum correlation formalized by Schrödinger, describes the ability of one party (Alice) to remotely change, or "steer," the state of a distant party (Bob) by performing local measurements exploiting shared entanglement [?]. A major practical application of quantum steering is the preparation of specific quantum states, known as Measurement-Induced Quantum Steering (MIQS), a protocol designed for Quantum State Preparation (QSP) to prepare a desired target state irrespective of the initial (mixed) state [?]. Quantum Steering is vital for providing security in one-sided

device-independent Quantum Key Distribution (1SDI QKD) protocols, where the degree of Gaussian steerability can quantify the guaranteed key rate achievable [?]. Quantum steering is also closely linked to quantum teleportation, any gaussian state capable of performing secure teleportation must be two way steerable [?]. Quantum steering also plays an important role in quantum-enhanced metrology [?].

2.8.3 Applications of Waveguides

Coupled waveguide structures are critical foundational components investigated for their viability as basic units of quantum circuits in integrated photonic applications. Discrete optical systems like these are known to be extremely efficient in manipulating the flow of light. Waveguides have been used to implement key operations for quantum computation, such as the realization of a CNOT gate on a single silicon chip using coupled silica waveguides. They also facilitate the generation of multimode interferometers that can be used to generate arbitrary quantum circuits and form the basic element for constructing the quantum circuit for Shor's Quantum Factoring Algorithm on a photonic chip. Beyond computation, waveguide arrays are useful for studying coherent phenomena such as quantum walk effects and the discrete analogue of the Talbot effect [?].